

Hypothesis Testing and Non-parametric Testing

From Formulating Claims to Making Decisions

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MAI 600: Mathematics and Statistics for AI

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Lecture Outline

Foundations of Hypothesis Testing

Parametric Tests

Non-parametric Testing

Summary

A Motivating Example

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- ▶ What if you get **7 heads**?

The Central Question

How do we decide whether our assumption is correct based on the evidence?

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Why Study This for AI?

- ▶ **A/B Testing:** Is a new recommendation algorithm in an app *statistically better* than the old one? Hypothesis testing provides the framework to answer this definitively.
- ▶ **Model Comparison:** Does a new, complex neural network provide a *significant improvement* in accuracy over a simpler baseline model, or is the improvement just due to random chance?
- ▶ **Feature Selection:** The Chi-Square test can determine if there's a significant association between a categorical feature (e.g., 'user_country') and the target variable (e.g., 'purchased_subscription'), helping us build more effective models.

The Two Competing Hypotheses

Hypothesis testing is like a statistical court case. We start by assuming the "status quo" is true.

Null Hypothesis (H_0)

The "status quo" assumption. It is the claim we seek evidence *against*.

- ▶ *Example:* A new drug's cure rate is the same as the old drug's rate.

Alternative Hypothesis (H_1 or H_a)

It is the claim we are trying to support with data.

- ▶ *Example:* The new drug's cure rate is higher than the old drug's.

Our Goal: To determine if our sample data provides enough evidence to *reject* the null hypothesis.

Type I and Type II Errors

No decision is ever 100% certain. We can make two kinds of mistakes.

Type I Error (α): Rejecting a true null hypothesis (false positive). The probability of this error is the **significance level** α .

Type II Error (β): Failing to reject a false null hypothesis (false negative).

Table: The Tradeoff in Hypothesis Decisions

Decision	H_0 is True	H_0 is False
Reject H_0	Type I Error (α)	Correct Decision
Fail to Reject H_0	Correct Decision	Type II Error (β)

The Hypothesis Testing Workflow

1. **Formulate Hypotheses & Set Criteria:** State the null (H_0) and alternative (H_1) hypotheses. Choose a significance level α (e.g., 0.05) as your evidence threshold.
2. **Calculate the Test Statistic:** Compute the test statistic from the sample that measures its deviation from the mean.
3. **Determine the p-value:** The p-value is the probability of observing your test statistic, *assuming the null hypothesis (H_0) is true*.
 - ▶ A small p-value means your observed data is very unlikely if the null hypothesis were true.
4. **Make a Statistical Decision:** Compare the p-value to α .

The Decision Rule

If the p-value is low, the null must go.

- ▶ If **p-value** $\leq \alpha$, we **reject** H_0 . The result is statistically significant.
- ▶ If **p-value** $> \alpha$, there is not enough evidence to reject H_0 .

Hypothesis Testing for Proportion: Example

Problem

An e-commerce site's current button design has a 10% click-through rate (CTR). They test a new design on 500 users, and 65 click it. Is the new design significantly better? (Use a significance level of $\alpha = 0.05$).

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- ▶ $H_0 : p \leq 0.10$ (The new design is not better than the old one).
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Step 2: Calculate Sample Proportion ($\hat{p}$)

$$\hat{p} = \frac{\text{successes}}{\text{trials}} = \frac{65}{500} = 0.13$$

Hypothesis Testing for Proportion: Calculation

Step 3: Calculate the Test Statistic (z-statistic)

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.13 - 0.10}{\sqrt{\frac{0.10(1-0.10)}{500}}} = \frac{0.03}{\sqrt{0.00018}} \approx 2.236$$

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Step 4: Make a Decision

- ▶ The p-value is the probability of observing a test statistic as extreme or more extreme than ours, *assuming H_0 is true*.
- ▶ For this right-tailed test, $p\text{-value} = P(Z > 2.236)$.
- ▶ From a Z-table or calculator, this probability is approximately 0.0127.

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Step 5: Conclusion

$$p\text{-value}(0.0127) < \alpha(0.05)$$

We **reject the null hypothesis**. There is statistically significant evidence to conclude that the new button design has a higher click-through rate.

Hypothesis Test for Mean (σ known, z-test)

Problem

A machine is designed to fill bags with 500g of coffee. The standard deviation of the fill weight is known to be $\sigma = 15\text{g}$. A quality check on a random sample of 40 bags finds a mean weight of 505g. Is there evidence at $\alpha = 0.05$ that the process is not meeting its target?

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1. Hypotheses (This is a two-tailed test)

- ▶ $H_0 : \mu = 500\text{g}$ (The process is on target).
- ▶ $H_1 : \mu \neq 500\text{g}$ (The process is off target).

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2. Test Statistic

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{505 - 500}{15/\sqrt{40}} \approx \frac{5}{2.372} \approx 2.108$$

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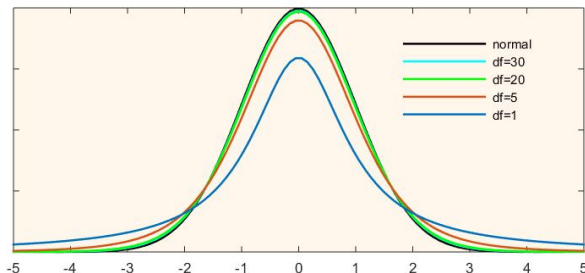
3. Decision & Conclusion

- ▶ For a two-tailed test, p-value = $2 \times P(Z > |2.108|) \approx 2 \times 0.0175 = 0.035$.
- ▶ Since the p-value (0.035) is less than α (0.05), we **reject** H_0 .

When Population Variance is Unknown: The t-test

In reality, we rarely know the true population standard deviation σ .

- ▶ When σ is unknown, we estimate it using the sample standard deviation, s .
- ▶ This adds extra uncertainty. To account for this, we use the **Student's t-distribution** instead of the Normal (Z) distribution.



Hypothesis Test for Mean (σ unknown, t-test)

Problem

A researcher tests a new learning algorithm. A sample of 10 students using it achieves a mean score of 85 with a sample standard deviation of 8. The historical average score is 80. Is the new algorithm significantly better ($\alpha = 0.05$)?

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2. Test Statistic (t-statistic)

- ▶ Degrees of Freedom: $df = n - 1 = 10 - 1 = 9$.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{85 - 80}{8/\sqrt{10}} \approx \frac{5}{2.53} \approx 1.976$$

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3. Decision & Conclusion

- ▶ Using a t-distribution calculator for a one-tailed test with $df = 9$, the p-value for $t = 1.976$ is approximately 0.04.
- ▶ Since the p-value (0.04) is less than α (0.05), we **reject** H_0 .

Chi-Square (χ^2) Test for Independence

Purpose

To determine if there is a statistically significant association between two **categorical variables**.

The Core Idea: Compare the *Observed* frequencies in our sample data to the frequencies we would *Expect* if the two variables were truly independent.

Hypotheses:

- ▶ H_0 : The two variables are independent (no association).
- ▶ H_1 : The two variables are not independent (they are associated).

Chi-Square Test: Example

Problem

An AI company wants to know if there is a relationship between a user's chosen operating system and their preferred web browser.

Observed Frequencies (O):

OS	Chrome	Firefox	Safari	Row Total
Windows	150	60	30	240
Mac	40	20	80	140
Linux	25	20	5	50
Col Total	215	100	115	430

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Calculate Expected Frequencies (E):

$$E = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Grand Total}}$$

Example for (Windows, Chrome): $E = \frac{240 \times 215}{430} = 120.0$

Chi-Square Test: Calculation & Conclusion

Expected Frequencies (E):

OS	Chrome	Firefox	Safari
Windows	120.0	55.8	64.2
Mac	70.0	32.6	37.4
Linux	25.0	11.6	13.4

1. Test Statistic (χ^2)

$$\chi^2 = \sum \frac{(O - E)^2}{E} = \frac{(150 - 120)^2}{120} + \dots + \frac{(5 - 13.4)^2}{13.4} \approx 95.3$$

2. Degrees of Freedom:

$$df = (\text{rows} - 1)(\text{cols} - 1) = (3 - 1)(3 - 1) = 4.$$

3. Decision & Conclusion

- ▶ p-value $\approx 1.25 \times 10^{-19}$ (This is virtually zero).
- ▶ We **reject** H_0 . There is a very strong association.

Why Non-parametric Tests?

Problem

Parametric tests like the t-test have a key assumption: the data is drawn from a normally distributed population. What if this is not true?

When data is:

- ▶ Heavily skewed
- ▶ Has significant outliers
- ▶ Is ordinal (ranked) rather than interval

...the assumptions of parametric tests are violated, and their results can be unreliable.

Solution

Non-parametric tests are "distribution-free." They do not rely on assumptions about the underlying population distribution. They often use ranks or signs of the data instead of the actual values.

Sign Test for a Single Median

Purpose

A simple test for a claim about the **median** of a population. It is extremely robust to outliers.

Problem

A company claims the median satisfaction score for their product is 7. A sample of 11 users gives the scores: {8, 9, 6, 10, 8, 9, 5, 4, 9, 8, 9}. Test this claim at $\alpha = 0.05$.

1. Hypotheses: $H_0 : \text{Median} = 7$, $H_1 : \text{Median} \neq 7$.

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1. **Hypotheses:** $H_0 : \text{Median} = 7$, $H_1 : \text{Median} \neq 7$.
2. **Assign Signs** (Compare each score to the median of 7)
 - ▶ Scores: {8, 9, 6, 10, 8, 9, 5, 4, 9, 8, 9}
 - ▶ Signs: {+, +, -, +, +, +, -, -, +, +, +}

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 - ▶ Scores: $\{8, 9, 6, 10, 8, 9, 5, 4, 9, 8, 9\}$
 - ▶ Signs: $\{+, +, -, +, +, +, -, -, +, +, +\}$
3. **Test Statistic:** The number of the less frequent sign.
 - ▶ Number of '+': 8
 - ▶ Number of '-': 3
 - ▶ The test statistic is $S = 3$. Our sample size is $n = 11$.

4. Find the p-value

- ▶ Under H_0 , the number of '+' or '-' signs should follow a Binomial distribution with $n = 11$ and $p = 0.5$.
- ▶ The p-value for a two-tailed test is $2 \times P(X \leq S)$.
- ▶ $\text{p-value} = 2 \times P(X \leq 3 \mid n = 11, p = 0.5)$.
- ▶ From a binomial calculator:
$$P(X \leq 3) = P(0) + P(1) + P(2) + P(3) \approx 0.113$$
- ▶ $\text{p-value} = 2 \times 0.113 = 0.226$.

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5. Conclusion

- ▶ Since the p-value (0.226) is greater than α (0.05), we **fail to reject the null hypothesis**.
- ▶ There is not enough evidence to conclude that the median satisfaction score is different from 7.

Wilcoxon Signed-Rank Test

Purpose

Also tests the population median, but it is more powerful than the Sign Test because it considers the **magnitude** of the differences from the median, not just their sign. It assumes the population distribution is symmetric.

Problem

Let's re-examine the satisfaction score data: {8, 9, 6, 10, 8, 9, 5, 4, 9, 8, 9}. H_0 : Median = 7, H_1 : Median $\neq$ 7.

Wilcoxon Signed-Rank Test: Cont

1. Calculate Differences & Ranks

Score	Difference	Difference	Rank
8	+1	1	2.5
9	+2	2	6.5
6	-1	1	2.5
10	+3	3	8.5
8	+1	1	2.5
9	+2	2	6.5
5	-2	2	6.5
4	-3	3	8.5
9	+2	2	6.5
8	+1	1	2.5
9	+2	2	6.5

Wilcoxon Signed-Rank Test: Cont

Key Rules for Ranking

- ▶ **Order:** Rank based on the absolute value of the differences, from smallest to largest.
- ▶ **Ties:** If multiple values are tied, they each receive the *average* of the rank positions they would have occupied.
- ▶ **Zeros:** If any difference is exactly zero, that data point is removed from the analysis, and the sample size n is reduced accordingly.

Calculating the Rank of 2.5

The four smallest absolute differences are all '1'. They occupy the first four rank positions (1, 2, 3, and 4). Their shared rank is the average of these positions:

$$\frac{1 + 2 + 3 + 4}{4} = \frac{10}{4} = 2.5$$

Wilcoxon Signed-Rank Test: Conclusion

2. Sum the Ranks

- ▶ Sum the ranks corresponding to the original positive differences (W^+). $W^+ = 2.5 + 6.5 + 8.5 + 2.5 + 6.5 + 6.5 + 2.5 + 6.5 = 42$
- ▶ Sum the ranks corresponding to the original negative differences (W^-). $W^- = 2.5 + 6.5 + 8.5 = 17.5$

3. Test Statistic: The test statistic W is the minimum of W^+ and W^- .

$$W = \min(42, 17.5) = 17.5$$

4. Decision and Conclusion

- ▶ $p\text{-value} = 0.167$
- ▶ We **fail to reject** H_0 .

- ▶ **Hypothesis Testing** is a formal procedure for using sample data to make decisions about claims regarding a population.
- ▶ Decisions involve choosing between a Null Hypothesis (H_0) and an Alternative Hypothesis (H_1).
- ▶ **Parametric Tests** (z-test, t-test, χ^2 -test) are powerful tools for specific data types (proportions, means, categorical) but rely on distributional assumptions like normality.
- ▶ **Non-parametric Tests** (Sign Test, Wilcoxon Signed-Rank Test) are robust, "distribution-free" alternatives used when the assumptions of parametric tests are not met.

Thank you. Questions?